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Hand-Grip Strength: Normative Reference Values and Equations for Individuals 18 to 85 Years of Age Residing in the United States

Published research provides extensive support for the measurement of hand-grip strength. That research shows grip strength to be (1) an acceptable indicator of overall muscle strength⁵; (2) impaired in various musculoskeletal,^{1,3} neuromuscular,¹⁰ and cardiovascular/pulmonary conditions^{3,4}; (3) fundamental to the diagnosis of frailty¹⁸ and sarcopenia¹²; and (4) prognostic of untoward outcomes such as mortality and future disability.⁴ Grip strength has been recommended for routine use as a vital sign⁴ and as a screening tool in primary care²² and hospital practice.²¹

The identification of grip-strength impairments requires normative reference values to which an individual's grip-strength measurements can be compared. There are numerous peer-re-

viewed studies that provide such values for populations outside the United States. These populations include those of Great Britain,¹³ Germany,²⁰ Spain,²³ Australia,²⁴ Canada,⁴¹ Korea,³⁵ Taiwan,⁴² and Japan.³³ Studies^{25,26,28} that provide grip-strength reference values have also been conducted in the United States. Unfortunately, most of these studies^{25,26} were completed before 1986 and used small convenience samples. Ideally, normative reference values should have been obtained within the past 15 to 20 years³⁸ from a population-representative sample.³¹

There are 2 large-scale, relatively recent studies involving the population of the United States that provide grip-strength data over the lifespan and so might be used to generate normative reference values. One study, the National Health and Nutrition Examination Survey (NHANES),¹¹ is an ongoing research program conducted by the US Centers for Disease Control and Prevention and National Center for Health Statistics to assess the health and nutritional status of adults and children in the United States and to track changes over time. Perna et al²⁸ reported grip-strength estimates using

• **BACKGROUND:** Hand-grip strength is an indicator of overall strength and a predictor of important outcomes. Up-to-date, population-specific reference values for measurements of grip strength are needed to properly interpret strength outcomes.

• **OBJECTIVES:** To provide population-based grip-strength reference values and equations for US residents 18 to 85 years of age.

• **METHODS:** Hand-grip data from 1232 participants 18 to 85 years of age were extracted from the database of the 2011 normative phase of the US National Institutes of Health Toolbox project in this cross-sectional study. Descriptive reference values and equations were derived from the data.

• **RESULTS:** The authors present grip-strength reference values using summary statistics

(mean, standard deviation, and percentile). The mean grip strength ranged from 49.7 kg for the dominant hand of men 25 to 29 years of age to 18.7 kg for the nondominant hand of women 75 to 79 years of age. The researchers also present reference regression equations for the dominant and nondominant sides of men and women. The explanatory variables in the equations are age, height, and weight.

• **CONCLUSION:** The normative reference values and equations provided in this study may serve as a guide for interpreting grip-strength measurements obtained from tested individuals. *J Orthop Sports Phys Ther* 2018;48(9):685-693. Epub 23 May 2018. doi:10.2519/jospt.2018.7851

• **KEY WORDS:** grip, hand, hand strength, NIH Toolbox (health care), norm

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2011-2012 NHANES data, but the authors looked at the strength of both hands combined. The other study is the US National Institutes of Health (NIH) Toolbox.¹⁹ It incorporates a multidimensional set of brief, royalty-free measures for assessing emotional, cognitive, sensory, and motor function in people aged 3 to 85 years. Data were collected for the normative phase of the NIH Toolbox project in 2011,² but up to this point, normative data have only been published in the peer-reviewed literature for individuals 3 to 17 years of age.⁸

The authors of the present study compared grip-strength outcomes and procedures of the NHANES and NIH Toolbox studies.⁹ Stratified grip-strength values derived from the data of the 2 studies were sometimes comparable, but the instrumentation and procedures used in these studies differed considerably; those of the NIH Toolbox were much more consistent with the recommendations of the American Society of Hand Therapists¹⁷ and Roberts et al.³² Those recommendations stipulate that the second handle position of the dynamometer be used when evaluating grip strength. Patients should be seated with the shoulder adducted and neutrally rotated, the elbow flexed at 90°, and the forearm and wrist in neutral position. Thus, data from the NIH Toolbox study were used to address the 2 purposes of the present study: (1) to provide US population-based grip-strength reference values for individuals 18 to 85 years of age, and (2) to develop normative regression equations for grip strength.

METHODS

NIH Toolbox Project

THE NIH TOOLBOX PROJECT IS A 4-year study designed to develop an assessment battery that provides a standard set of royalty-free, brief, and comprehensive assessment tools for use by researchers and clinicians in a variety of settings, with a particular emphasis on measuring outcomes in longitudinal, epidemiological, and prevention or intervention clinical research across the life-

span of individuals 3 to 85 years of age. Grip strength was identified in 2007 as a core construct for inclusion in the NIH Toolbox motor domain, based on literature and instrument database reviews, requests for information, consensus meetings, and expert interviews.

The NIH Toolbox team completed a validation study in 2009 of 340 healthy participants.^{7,30,39,40} Using the standardized testing protocol, the validation study assessed the score range across age groups (3 to 85 years of age), test-retest reliability (intraclass correlation coefficient = 0.98), and concurrent and known-groups validity of grip strength.

When the final measures were recommended in all domains, the NIH Toolbox management core conducted a large national norming phase in 2011 to allow for normative comparison on each measure. All of the NIH Toolbox measures were administered to a total of 4859 participants, aged 3 to 85 years and representative of the US population. Hence, participants took the grip-strength test as part of the entire NIH Toolbox battery.

Participants were randomized into 4 different data-collection orders (ie, 4 study arms): cognition, motor, emotion, and sensation domains. Sampling was stratified by age, sex, and primary language (English, Spanish). Within each age stratum, target quotas were set according to the US population distribution of race, ethnicity, and level of education (parents' education for children). Preliminary results suggested no order effect for grip strength ($P = .621$). Detailed norming plans for the NIH Toolbox have been described.²

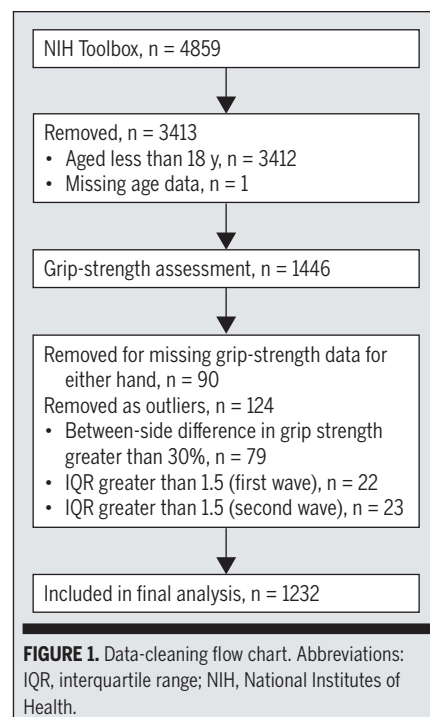
Participants

Grip-strength data collected in the cross-sectional normative phase of the NIH Toolbox project from August to November 2011 were analyzed. Inclusion required that an individual was (1) community dwelling and noninstitutionalized, (2) aged 3 to 85 years, (3) capable of following test instructions (English or Spanish), and (4) able to give informed

consent or, in the case of children, to give assent with accompanying informed consent by proxy (ie, parent/guardian). Data were collected at 10 sites, primarily urban and suburban (Atlanta, Chicago-Oak Brook, Cincinnati, Columbus, Dallas, Los Angeles, Minneapolis, Philadelphia, Phoenix, and St Louis).

The norms of hand-grip strength for participants 3 to 17 years of age have been published.⁸ Here, we only analyzed the adult data. From the initial 4859 data records, 3412 were removed because the age was younger than 18 years old. Subsequently, 91 were removed because of missing age or grip-strength data. An additional 124 were excluded as outliers because their grip-strength values were outside the 1.5 interquartile range of the mean for other participants of the same sex and age group, or because the between-side difference in grip strength was larger than 30%. **FIGURE 1** illustrates the data-cleaning process.

Data from 1232 remaining participants (aged between 18 and 85 years) were included in the final analysis. Of these, 449 (36.4%) were men and 783 (63.6%) were women. By self-report, 1151 (93.4%) participants in the sample were right-hand



dominant. The majority of the sample was white (77.8%, $n = 958$). Other races were represented as black or African American (12.5%, $n = 154$), American Indian or Alaska native (3.1%, $n = 38$), and Asian (2.6%, $n = 32$). Four hundred fifty-seven (37.1%) were Hispanic or Latino, while 774 (62.8%) were not. Regarding test instructions, 861 (69.9%) participants received the instructions in English, 276 (22.4%) in Spanish, and 95 (7.7%) in bilingual (English and Spanish) format.

The Institutional Review Board of Northwestern University approved the data-collection procedures. All participants consented to the study after being informed about the study's purpose and procedures. Additional Institutional Review Board approval was waived, as this was a secondary data analysis free of personal identifiers.

Grip-Strength Test Protocol

The test administrator's manual, which includes detailed descriptions of the protocol, and demonstration videos are available on the HealthMeasures website (<http://www.healthmeasures.net>) in both English (119 pages) and Spanish (156 pages).³⁶ Briefly, a calibrated digital Jamar dynamometer (Performance Health, Warrenville, IL), with its handle in the second position, was used. Participants were instructed to squeeze the dynamometer as hard as they could while seated in an upright posture, with arms by their sides, elbows flexed to 90°, and forearms in a neutral position.

The test administrator showed the dynamometer to the participant and said, "We will use this machine to measure how strong your hands are. You will squeeze the handle 2 times (1 practice and 1 test trial) with each hand while your arm is against your side and your elbow is bent like this (90°). The handle won't move, but the machine will show how hard you squeezed. See? (Show force measurement.) Do you have any questions?"

During the practice trial, the test administrator said, "Let's practice. First, we'll use this (dominant hand, your right

or left) hand. When I say, 'Squeeze,' I want you to squeeze the handle hard, but not as hard as you can. [Support the dynamometer during testing.] Ready? 3, 2, 1, squeeze." After 3 to 4 seconds, the test administrator said, "Stop."

During the test trial, the test administrator said, "Now we're going to test each hand, beginning with this [Point and say, 'Your right or left'] hand. When I say, 'Squeeze,' this time I want you to squeeze the handle as hard as you can." The test administrator supported the dynamometer during testing.

Due to time constraints in the norming study of the entire NIH Toolbox battery, for grip-strength testing, a single submaximal practice trial was conducted for each hand, followed by a 30-second rest and a single maximal trial of 3 to 4 seconds for each hand. Participants were encouraged by the examiner, who chanted, "Harder, harder, harder." The force associated with the maximal trial was documented in pounds and later converted to kilograms.

Statistical Analysis

All analyses were performed using SPSS Statistics for Windows Version 23.0 (IBM Corporation, Armonk, NY). To compare hand-grip measures, participants were categorized into 13 age groups: 18 to 24, 25 to 29, 30 to 34, 35 to 39, 40 to 44, 45 to 49, 50 to 54, 55 to 59, 60 to 64, 65 to 69, 70 to 74, 75 to 79, and 80 to 85 years. A 2-by-2-by-13 (sex by side by age group) general linear model (GLM) analysis was used to test the hypothesis that these 3 independent variables should be used to stratify grip-strength measurements. For independent variables found to have a significant main or interactive effect on grip strength, pairwise post hoc comparisons were conducted using the GLM. The effect of other potentially explanatory variables on grip strength was also examined using the GLM. These variables included body mass index (BMI), ethnicity, education, language, and handedness. Descriptive statistics of grip strength were tabulated for relevant strata.

Prior to generating the regression equations, scatter plots and Pearson correlation coefficients were used to explore the relationships between grip strength and independent variables of interest, including height, weight, BMI, and age. Note that all 4 variables are continuous variables. Mukaka's²⁷ guideline was used to interpret correlation coefficients in medical research: greater than 0.9 (very high), 0.7 to 0.9 (high), 0.5 to 0.7 (moderate), 0.3 to 0.5 (low), and less than 0.3 (negligible). Data were checked for assumptions of normality, linearity, and multicollinearity. Four forward multiple linear regression analyses (1 for each sex and side category) were used to generate explanatory equations for grip strength.

The initial explanatory variables included age, age squared, age cubed, height, weight, and BMI. Age squared and age cubed were used because the scatter plots between grip strength and age showed a nonlinear relationship. Data points with Cook's distance greater than $4/(n - 2)$, where n is the sample size, were considered as influential points³⁷ and were removed from the regression analysis. Model fit was inspected using the overall F test for the regression model, individual t tests for each regression coefficient, and adjusted R^2 . Based on the numerous hypothesis tests conducted and a desire to reduce the risk of type I error, a significance level of $P < .005$ was adopted as an indicator of statistical significance.

RESULTS

FIGURE 2 ILLUSTRATES THE TRAJECTORIES of grip-strength values from individuals 18 to 85 years of age. The age group is on the x -axis, while the mean grip strength in kilograms is on the y -axis and is presented by sex and side. General linear model results verified that grip strength differed significantly ($P < .001$) by sex (with men stronger than women), by side (with the dominant side stronger than the nondominant side), and by age group (with younger adults stronger than

[RESEARCH REPORT]

older adults). No significant interaction effects were found. **TABLE 1** summarizes grip strength (kilograms) by sex, side, and age group. To assist in clinical interpretation of grip-strength data, the authors also reported the summary of hand-grip strength measures in pounds (**APPENDIX A**, available at www.jospt.org) and presented box plots of grip strength by sex and side in kilograms (**FIGURE 3**) and pounds (**APPENDIX B**, available at www.jospt.org).

Pairwise comparisons showed that grip strength for participants aged 18 to 24 years was not statistically different

from participants in nearby age groups (25-59 years) but was significantly greater than that of participants in age groups of 60 years and older ($P < .001$). Further GLM analysis showed that Hispanic or Latino participants had lower grip strength than participants classified as not Hispanic or Latino ($P < .001$). Participants with a high school (HS) diploma (ie, HS graduate) and a higher degree (more than HS) were stronger than participants who did not finish secondary education (less than HS) ($P < .001$). Participants who were given instructions in English showed higher

grip strength than participants who were given instructions in Spanish only ($P < .001$); however, the differences were not significant between bilingual (English and Spanish) and Spanish only ($P = .049-.720$). At a significance level of .005, grip-strength values were not statistically different by BMI category ($P = .045$) or by handedness ($P = .007$). **TABLE 2** summarizes these GLM comparisons.

Among demographic variables, correlations with grip strength were significant and moderate for height ($r = 0.61$) and significant and low for weight ($r =$

TABLE 1

**SUMMARY OF HAND-GRIP STRENGTH MEASUREMENTS
BY SIDE, SEX, AND AGE-GROUP STRATA***

Hand/Sex/Age, y	Height, m	Weight, kg	Strength, kg	Percentile				
				10	25	50	75	90
Dominant								
Male								
18-24 (n = 36)	1.81 ± 0.08	82.0 ± 16.9	47.0 ± 8.1	36.2	41.4	47.8	51.2	57.9
25-29 (n = 35)	1.78 ± 0.07	86.4 ± 21.9	49.7 ± 11.6	33.7	43.3	49.3	59.4	66.2
30-34 (n = 29)	1.75 ± 0.06	91.0 ± 17.9	46.5 ± 12.1	31.2	36.4	46.1	56.4	63.1
35-39 (n = 41)	1.77 ± 0.07	92.5 ± 22.1	47.1 ± 11.9	30.3	39.7	50.1	54.3	60.8
40-44 (n = 47)	1.75 ± 0.07	90.0 ± 19.6	46.7 ± 11.7	34.3	39.9	45.9	54.4	63.1
45-49 (n = 32)	1.73 ± 0.06	89.1 ± 17.9	42.8 ± 10.9	31.1	35.8	40.7	48.2	59.2
50-54 (n = 46)	1.78 ± 0.08	93.8 ± 17.2	44.0 ± 10.3	30.4	39.0	44.8	52.3	56.7
55-59 (n = 27)	1.77 ± 0.08	92.3 ± 22.4	40.7 ± 10.4	28.2	32.4	38.7	47.8	56.3
60-64 (n = 33)	1.77 ± 0.08	90.3 ± 13.4	38.4 ± 10.3	23.3	30.4	40.3	44.9	52.5
65-69 (n = 22)	1.74 ± 0.08	86.2 ± 17.9	36.8 ± 10.5	17.8	31.5	36.6	45.8	50.1
70-74 (n = 39)	1.75 ± 0.08	88.3 ± 18.0	34.7 ± 9.0	16.7	29.3	36.3	41.2	45.6
75-79 (n = 24)	1.76 ± 0.08	86.2 ± 14.0	32.7 ± 10.1	18.4	25.9	33.5	36.6	43.5
80-85 (n = 38)	1.75 ± 0.08	81.1 ± 13.0	28.1 ± 9.1	15.6	21.5	29.5	34.6	38.2
Female								
18-24 (n = 54)	1.61 ± 0.07	72.3 ± 21.2	28.1 ± 7.1	17.6	22.4	28.4	33.8	38.0
25-29 (n = 102)	1.61 ± 0.07	73.3 ± 20.1	29.6 ± 7.0	20.2	25.4	29.6	33.6	39.7
30-34 (n = 109)	1.63 ± 0.07	76.1 ± 19.6	28.9 ± 6.2	20.5	23.9	29.8	33.0	37.1
35-39 (n = 90)	1.62 ± 0.07	75.2 ± 17.4	29.2 ± 6.2	20.0	24.5	30.3	33.0	38.0
40-44 (n = 88)	1.63 ± 0.07	75.9 ± 18.4	29.9 ± 6.2	22.8	26.5	30.4	33.8	37.4
45-49 (n = 52)	1.63 ± 0.08	79.7 ± 19.1	28.8 ± 7.2	17.7	25.2	28.7	34.4	37.6
50-54 (n = 65)	1.63 ± 0.07	75.6 ± 16.0	28.2 ± 6.3	19.7	24.6	28.2	32.7	35.2
55-59 (n = 30)	1.62 ± 0.07	76.6 ± 16.2	25.1 ± 6.2	16.9	20.7	24.1	30.2	32.2
60-64 (n = 58)	1.62 ± 0.07	76.7 ± 17.4	23.6 ± 6.5	15.9	19.2	24.4	28.1	31.8
65-69 (n = 29)	1.62 ± 0.07	80.0 ± 21.5	22.1 ± 6.6	11.7	19.3	22.2	25.0	31.2
70-74 (n = 43)	1.60 ± 0.07	77.4 ± 18.8	21.5 ± 5.1	15.2	19.5	22.5	23.9	27.5
75-79 (n = 17)	1.58 ± 0.08	66.7 ± 10.4	19.6 ± 6.0	12.6	15.7	18.2	22.4	27.8
80-85 (n = 46)	1.60 ± 0.06	70.0 ± 11.3	19.9 ± 4.4	14.5	16.6	19.5	21.8	27.0

Table continues on page 689.

0.34-0.36) and age ($r = -0.26$ to -0.27). In multiple regression analysis (forward selection), 89 participants with Cook's distance greater than 0.00325 ($4/[1232 - 2]$) were considered as influential points and were removed. Regression analysis resulted in age cubed, height, and weight being selected as the best explanatory variables, with adjusted R^2 values ranging from 0.32 to 0.44. The overall F test was significant ($P < .001$), as were the t tests for the 3 regression coefficients ($P < .001$). **TABLE 3** provides normative regression equations for describing grip strength. **APPENDIX C** (available at www.jospt.org)

shows the scatter plots of the observed data and the predicted grip-strength values derived from the regression equations. Using the same existing data, the authors conducted a preliminary analysis examining the accuracy of the regression models. For men, mean $\pm$ SD differences between observed and predicted grip strength were -0.69 ± 9.6 and -0.62 ± 9.2 kg for dominant and nondominant hands, respectively. For women, mean $\pm$ SD differences between observed and predicted grip strength were -0.24 ± 6.1 and -0.30 ± 6.0 kg for dominant and nondominant hands, respectively.

DISCUSSION

THIS STUDY PRESENTS NORMATIVE reference values and equations for grip strength based on data obtained from the US population within the last 10 years. Notably, the data were acquired using procedures consistent with most published recommendations.^{17,32} Normative data provided in this study may be beneficial for diverse medical and ergonomic research, enabling clinicians and researchers to compare the grip-strength values in individuals with or without impairments to the reference

TABLE 1

SUMMARY OF HAND-GRIP STRENGTH MEASUREMENTS BY SIDE, SEX, AND AGE-GROUP STRATA* (CONTINUED)

Hand/Sex/Age, y	Height, m	Weight, kg	Strength, kg	Percentile				
				10	25	50	75	90
Nondominant								
Male								
18-24 (n = 36)	1.81 ± 0.08	82.0 ± 16.9	44.9 ± 7.8	35.7	38.0	44.5	50.4	55.2
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*Values are mean $\pm$ SD unless otherwise indicated.

TABLE 2

COMPARISONS OF HAND-GRIP STRENGTH MEASUREMENTS BY PARTICIPANT CHARACTERISTICS*

Variable	Men			Women		
	n	Dominant Hand	Nondominant Hand	n	Dominant Hand	Nondominant Hand
BMI, kg/m ²						
<18.5 (underweight)	1	40.8	38.2	13	25.2 ± 3.7	25.1 ± 3.9
18.5-24.9 (normal weight)	125	39.9 ± 11.5	38.7 ± 10.7	236	27.0 ± 6.7	25.8 ± 6.4
25-29.9 (overweight)	161	42.2 ± 11.8	40.4 ± 11.6	238	27.5 ± 6.8	26.5 ± 6.8
≥30 (obese)	141	43.5 ± 12.9	41.7 ± 12.2	257	27.1 ± 8.0	25.6 ± 7.6
Ethnicity [†]						
Not Hispanic or Latino	288	43.6 ± 11.8	41.9 ± 11.1	486	27.8 ± 7.1	26.7 ± 7.0
Hispanic or Latino	160	38.0 ± 12.2	36.5 ± 11.9	297	25.9 ± 7.0	24.4 ± 6.6
Education [†]						
Less than high school	102	36.4 ± 13.0	34.6 ± 12.5	156	25.1 ± 7.3	24.1 ± 7.2
High school graduate	104	43.2 ± 11.8	41.3 ± 10.7	168	26.1 ± 6.9	25.0 ± 6.6
More than high school	213	44.5 ± 10.6	42.8 ± 10.3	409	28.3 ± 7.0	27.0 ± 6.8
Language of administration [†]						
English only	320	43.8 ± 11.6	42.0 ± 11.1	541	27.8 ± 7.1	26.6 ± 6.9
Spanish only	97	34.9 ± 12.5	33.8 ± 12.3	179	25.0 ± 6.9	23.3 ± 6.4
Bilingual, English primary	10	41.4 ± 10.7	41.0 ± 10.4	18	28.2 ± 5.7	26.5 ± 5.2
Bilingual, Spanish primary	22	39.9 ± 9.8	37.5 ± 8.4	45	26.8 ± 7.6	26.0 ± 7.2
Handedness						
Right handed	411	41.4 ± 12.4	39.7 ± 11.8	740	27.0 ± 7.1	25.7 ± 6.9
Left handed	38	43.9 ± 10.3	43.0 ± 10.4	43	28.5 ± 7.0	28.3 ± 7.0

Abbreviation: BMI, body mass index.

*Values are mean ± SD kilograms unless otherwise indicated.

[†]The main effect was statistically significant ($P < .001$) for both dominant and nondominant hands. No main effect was statistically significant ($P < .001$) for 1 side only.

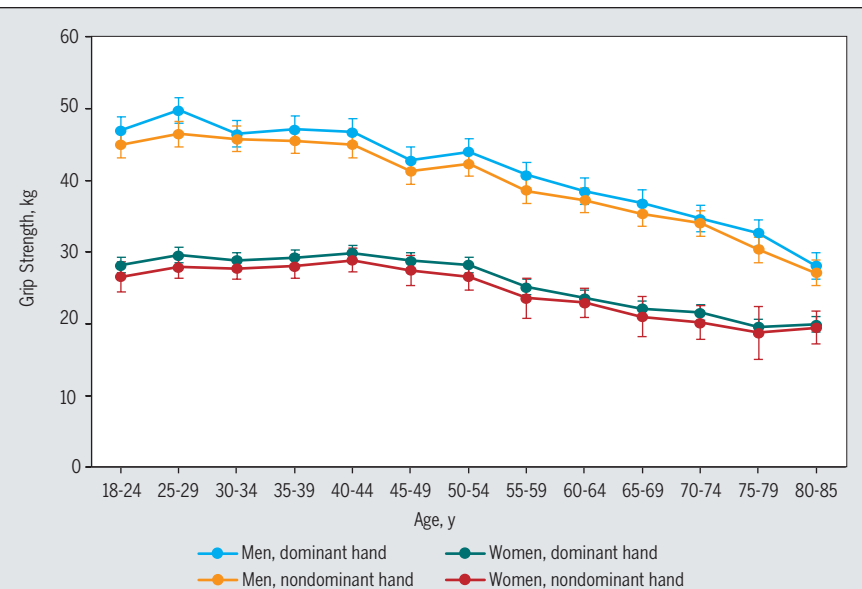


FIGURE 2. Trajectories of grip-strength measurements for individuals 18 to 85 years of age. The line indicates the average values of the hand-grip strength. The error bars indicate the 95% confidence interval for each age group.

values established based on the general, relatively healthy population. Clinicians and researchers could use grip strength to document progression of hand muscle strength and to provide feedback during the rehabilitation process with the knowledge of normal score ranges for individuals of similar age and sex.

Based on precedence and the results of this GLM analysis, the authors stratified normative reference values by side, sex, and age, with all of the age strata but 1 (18-24 years) limited to 5-year spans. In addition to means and standard deviations for each stratum, the authors provided percentiles of grip strength to assist clinicians and researchers in interpreting measurements obtained from tested individuals. The values presented are mostly less than, but within 1 standard deviation of, those provided by Dodds et al¹⁴

for participants in 12 British studies. This difference may be due to the use of maximum grip strength of the participants in the Dodds et al¹⁴ study, whereas the present study used the results of 1 maximum effort for each hand. However, it may be that the grip strength of the 2 populations is different. The values presented are also less than, but usually within the confidence intervals of, those derived by Bohannon et al⁶ through meta-analysis. Those values, however, were presented for the right and left sides rather than for the dominant and nondominant sides, as in this study.

All of the data consolidated by Bohannon et al⁶ and most of the data consolidated by Dodds et al¹⁴ were obtained before the data of the NIH Toolbox normative phase study. Previously collected grip-strength reference values may not be the best standard by which to interpret the performance of individuals tested more recently. This was highlighted in a study by Fain and Weatherford¹⁶ that compared the grip strength of young adults (millennials) with the normative reference values obtained over 30 years prior by Mathiowetz et al²⁵ and found the more recently tested individuals to be weaker.

Although reference values stratified according to significant explanatory variables are useful, they may not provide as specific a referent as a regression equation. Reference equations for grip strength have been reported before and tend to include age, sex, height, and weight as independent variables.¹⁵ Based on correlational and regression analysis, the authors of this study identified 3 variables for inclusion in reference equations (weight, height, and age cubed). Of these, only age was used to stratify reference values. Sex and side, which are not included in the regression equations, are addressed by providing separate equations for each sex and side stratum. Notably, BMI was not included in this study's equations. Although grip-strength values are sometimes interpreted in light of BMI,¹⁸ grip strength was found in this

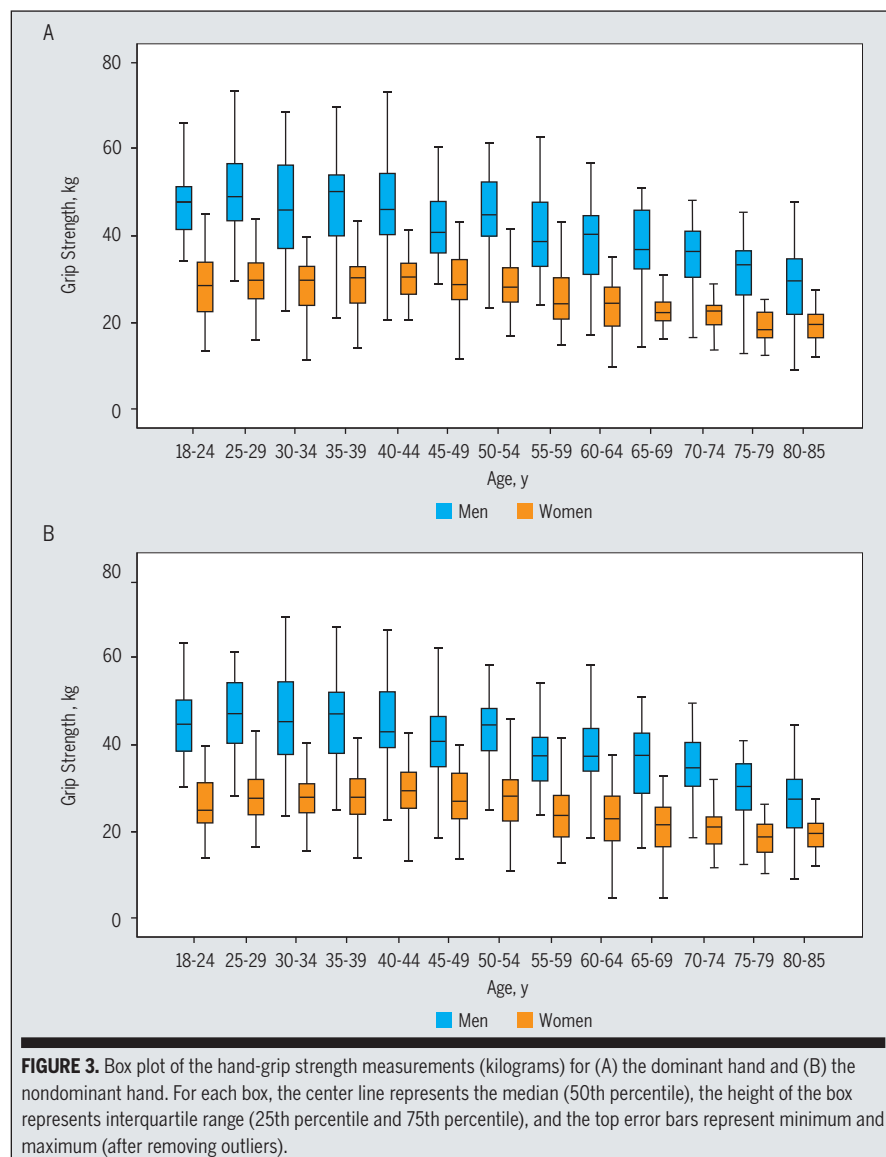


FIGURE 3. Box plot of the hand-grip strength measurements (kilograms) for (A) the dominant hand and (B) the nondominant hand. For each box, the center line represents the median (50th percentile), the height of the box represents interquartile range (25th percentile and 75th percentile), and the top error bars represent minimum and maximum (after removing outliers).

TABLE 3

**REFERENCE EQUATIONS OF GRIP STRENGTH
BY SEX AND HAND DOMINANCE**

Sex/Hand Dominance	Reference Equation	Adjusted R ²
Male		
Dominant	$-29.959 - 3.095E^{-05} \times (\text{age}^3) + 38.719 \times (\text{height}) + 0.113 \times (\text{weight})$	0.44
Nondominant	$-30.474 - 2.937E^{-05} \times (\text{age}^3) + 38.594 \times (\text{height}) + 0.099 \times (\text{weight})$	0.44
Female		
Dominant	$-22.717 - 1.920E^{-05} \times (\text{age}^3) + 30.360 \times (\text{height}) + 0.048 \times (\text{weight})$	0.36
Nondominant	$-21.292 - 1.776E^{-05} \times (\text{age}^3) + 28.995 \times (\text{height}) + 0.040 \times (\text{weight})$	0.32

*Age in years, height in meters, and weight in kilograms. E is the scientific notation. E⁻⁰⁵ indicates 10⁻⁵ = 0.00001. For example, the predicted grip strength of the dominant hand for a 42-year-old man with a height of 1.7 m and weight of 67.1 kg is $-29.959 - 3.095 \times 10^{-5} \times (42^3) + 38.719 \times (1.7) + 0.113 \times (67.1) = 41.2$ kg.

study to be correlated more highly with height than with BMI. Once the correlations of height and weight with grip strength were controlled, BMI made no further contribution to the explanation of grip strength.

A strength of this study is that it was based on the secondary analysis of data from a population-based study with rigorous sampling plans, data-collection procedures, and quality-control practices. An NIH Toolbox administration manual was provided for all sites to help with test standardization. In addition, examiners participating in the project were trained on standardized procedures prior to data collection. The authors of this study presented norms of hand-grip strength in the general population, following the inclusion criteria listed by the NIH Toolbox. Norms are statistics that describe the test performance of a predefined population (community-dwelling, noninstitutionalized adults in the present study). As normative data are not exactly the same as unaffected data, the data set may include individuals with comorbidities. Previous findings supported the 10% rule, which states that the dominant hand possesses a 10% greater grip strength than the nondominant hand in healthy subjects, and the differences are more noticeable for right-handed individuals.²⁹ Hence, as a conservative estimate, the researchers excluded 6.4% ($n = 79$) of grip-strength data, in which the between-side difference in grip strength was larger than 30%. Note that there were 54.3%, 84.9%, and 94.6% of data with side differences less than or equal to 10%, 20%, and 30%, respectively. The absence of data may reduce statistical power and the representativeness of the samples, and may cause bias in the analysis. However, side differences greater than 30% may indicate entry error or uncertain pathology. Because it is not feasible to check the primary source, the authors believe that removing side differences greater than 30% will improve the representation of the relatively healthy population, which facilitates the data interpretation. Ad-

ditionally, inspecting the side differences provided a way to check potential data-entry errors, as the authors did find subjects to have extreme hand-grip values of 105 kg in the dominant hand, but 1 kg in the nondominant hand.

Several limitations of the study must also be acknowledged. First, hand grip was measured as part of the entire set of NIH Toolbox measures. It is uncertain whether this may have affected the forces obtained. Second, although the sample was relatively large, its size was diminished by missing values (eg, demographics) and the exclusion of extreme values (outliers, data-entry errors). Ultimately, the sample was too small to justify stratification for factors (eg, ethnicity) determined to be relevant to grip strength.

Third, as this is a secondary data analysis, researchers were not in control of the data-collection procedures, and variables that may be important to grip strength (eg, hand size) were not addressed. Indeed, they could not be, as they were not included among the NIH Toolbox measures. Similarly, the authors do not have information about the number of examiners/raters and the inherent interrater reliability during the norming phase. Fourth, the researchers applied multiple regression analysis using the 4 independent variables that were available. To increase the predictive power of grip strength across the age span, more relevant variables (eg, hand size, ethnicity, comorbid burden, occupation) and more complicated statistical methods (such as Box-Cox power transformation and nonlinear regression models) could be explored. Last, future studies should endeavor to validate normative regression equations using a new and independent sample.

CONCLUSION

THE NORMATIVE REFERENCE VALUES provided herein can serve as a guide for interpreting grip-strength measurements obtained from adults in the United States. Reference equations can be used to the same end. ●

KEY POINTS

FINDINGS: This project provides US population-based, grip-strength reference values obtained from the US National Institutes of Health Toolbox project. These hand-grip values are summarized after stratification by side (ie, dominant and nondominant), sex, and age. Reference equations are provided.

IMPLICATIONS: The reference values provided can be used to interpret grip-strength measures obtained from adults in the United States.

CAUTION: The reference values and equations provided were obtained from a sample of insufficient size to justify additional stratification by race and other such factors.

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APPENDIX A

SUMMARY OF HAND-GRIP STRENGTH MEASUREMENTS
BY SIDE, SEX, AND AGE-GROUP STRATA

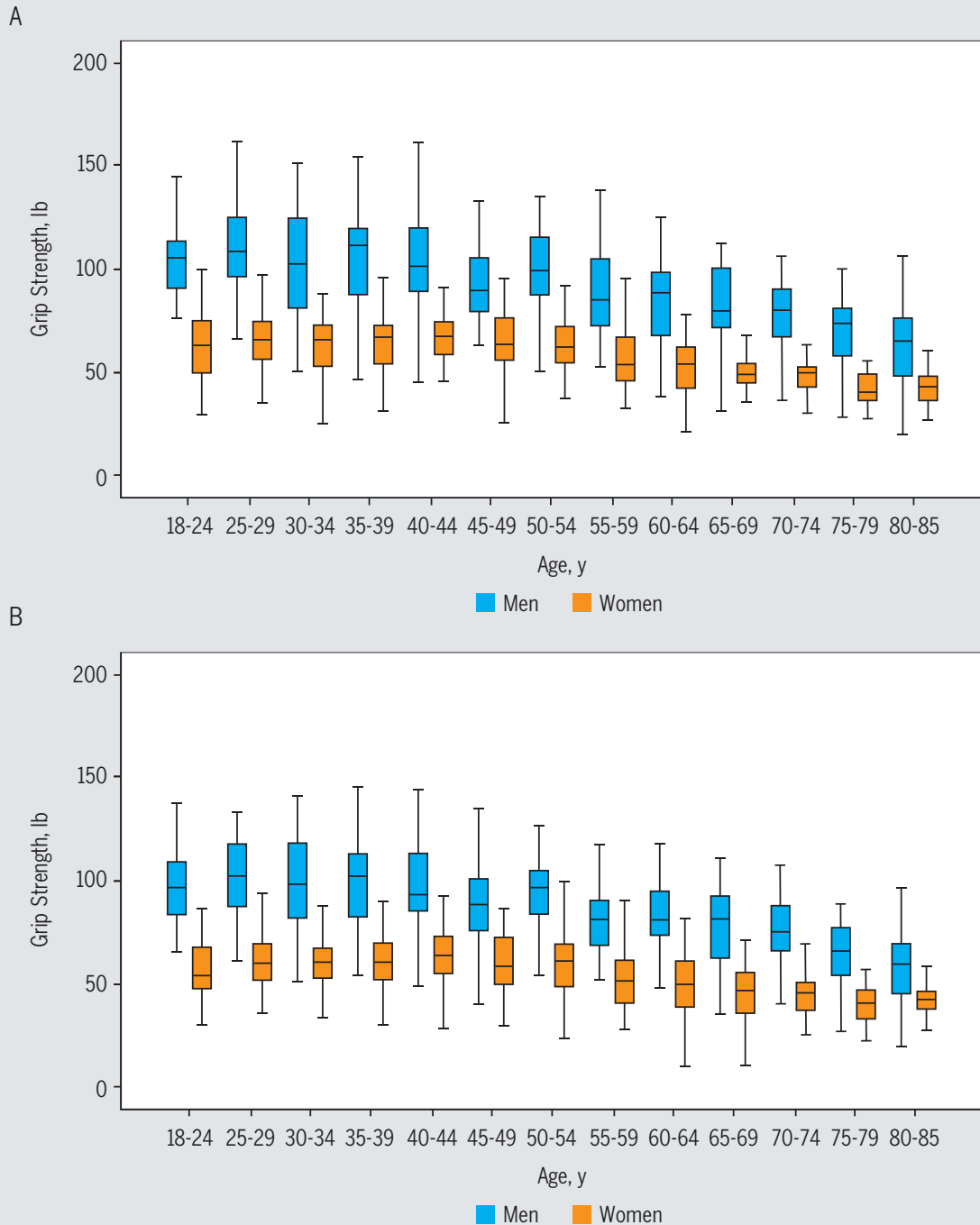
		Percentile				
Hand/Sex/Age, y	Strength, lb*	10	25	50	75	90
Dominant						
Male						
18-24 (n = 36)	103.6 ± 17.9	79.8	91.3	105.4	112.9	127.6
25-29 (n = 35)	109.6 ± 25.6	74.3	95.5	108.7	131.0	145.9
30-34 (n = 29)	102.5 ± 26.7	68.8	80.2	101.6	124.3	139.1
35-39 (n = 41)	103.8 ± 26.2	66.8	87.5	110.5	119.7	134.0
40-44 (n = 47)	103.0 ± 25.8	75.6	88.0	101.2	119.9	139.1
45-49 (n = 32)	94.4 ± 24.0	68.6	78.9	89.7	106.3	130.5
50-54 (n = 46)	97.0 ± 22.7	67.0	86.0	98.8	115.3	125.0
55-59 (n = 27)	89.7 ± 22.9	62.2	71.4	85.3	105.4	124.1
60-64 (n = 33)	84.7 ± 22.7	51.4	67.0	88.8	99.0	115.7
65-69 (n = 22)	81.1 ± 23.1	39.2	69.4	80.7	101.0	110.5
70-74 (n = 39)	76.5 ± 19.8	36.8	64.6	80.0	90.8	100.5
75-79 (n = 24)	72.1 ± 22.3	40.6	57.1	73.9	80.7	95.9
80-85 (n = 38)	61.9 ± 20.1	34.4	47.4	65.0	76.3	84.2
Female						
18-24 (n = 54)	61.9 ± 15.7	38.8	49.4	62.6	74.5	83.8
25-29 (n = 102)	65.3 ± 15.4	44.5	56.0	65.3	74.1	87.5
30-34 (n = 109)	63.7 ± 13.7	45.2	52.7	65.7	72.8	81.8
35-39 (n = 90)	64.4 ± 13.7	44.1	54.0	66.8	72.8	83.8
40-44 (n = 88)	65.9 ± 13.7	50.3	58.4	67.0	74.5	82.5
45-49 (n = 52)	63.5 ± 15.9	39.0	55.6	63.3	75.8	82.9
50-54 (n = 65)	62.2 ± 13.9	43.4	54.2	62.2	72.1	77.6
55-59 (n = 30)	55.3 ± 13.7	37.3	45.6	53.1	66.6	71.0
60-64 (n = 58)	52.0 ± 14.3	35.1	42.3	53.8	61.9	70.1
65-69 (n = 29)	48.7 ± 14.6	25.8	42.5	48.9	55.1	68.8
70-74 (n = 43)	47.4 ± 11.2	33.5	43.0	49.6	52.7	60.6
75-79 (n = 17)	43.2 ± 13.2	27.8	34.6	40.1	49.4	61.3
80-85 (n = 46)	43.9 ± 9.7	32.0	36.6	43.0	48.1	59.5
Nondominant						
Male						
18-24 (n = 36)	99.0 ± 17.2	78.7	83.8	98.1	111.1	121.7
25-29 (n = 35)	102.5 ± 21.2	68.6	86.9	104.1	124.3	132.1
30-34 (n = 29)	101.0 ± 24.9	62.8	81.6	99.2	123.9	132.5
35-39 (n = 41)	100.3 ± 24.3	75.6	82.7	104.1	115.3	129.6
40-44 (n = 47)	99.0 ± 25.8	70.8	85.5	94.1	115.5	135.4
45-49 (n = 32)	90.8 ± 22.0	65.3	75.8	89.1	102.5	127.2
50-54 (n = 46)	93.3 ± 23.4	59.7	84.4	97.7	107.4	121.5
55-59 (n = 27)	84.9 ± 21.2	60.4	67.7	82.0	93.7	121.9
60-64 (n = 33)	82.0 ± 20.1	51.6	70.3	81.8	98.8	108.7
65-69 (n = 22)	78.0 ± 22.7	38.1	61.7	82.7	94.8	105.8
70-74 (n = 39)	75.0 ± 20.9	45.2	65.9	76.1	89.5	100.8
75-79 (n = 24)	66.8 ± 21.8	32.0	54.0	66.6	79.4	88.6
80-85 (n = 38)	59.7 ± 20.7	31.3	44.1	60.2	70.5	88.2

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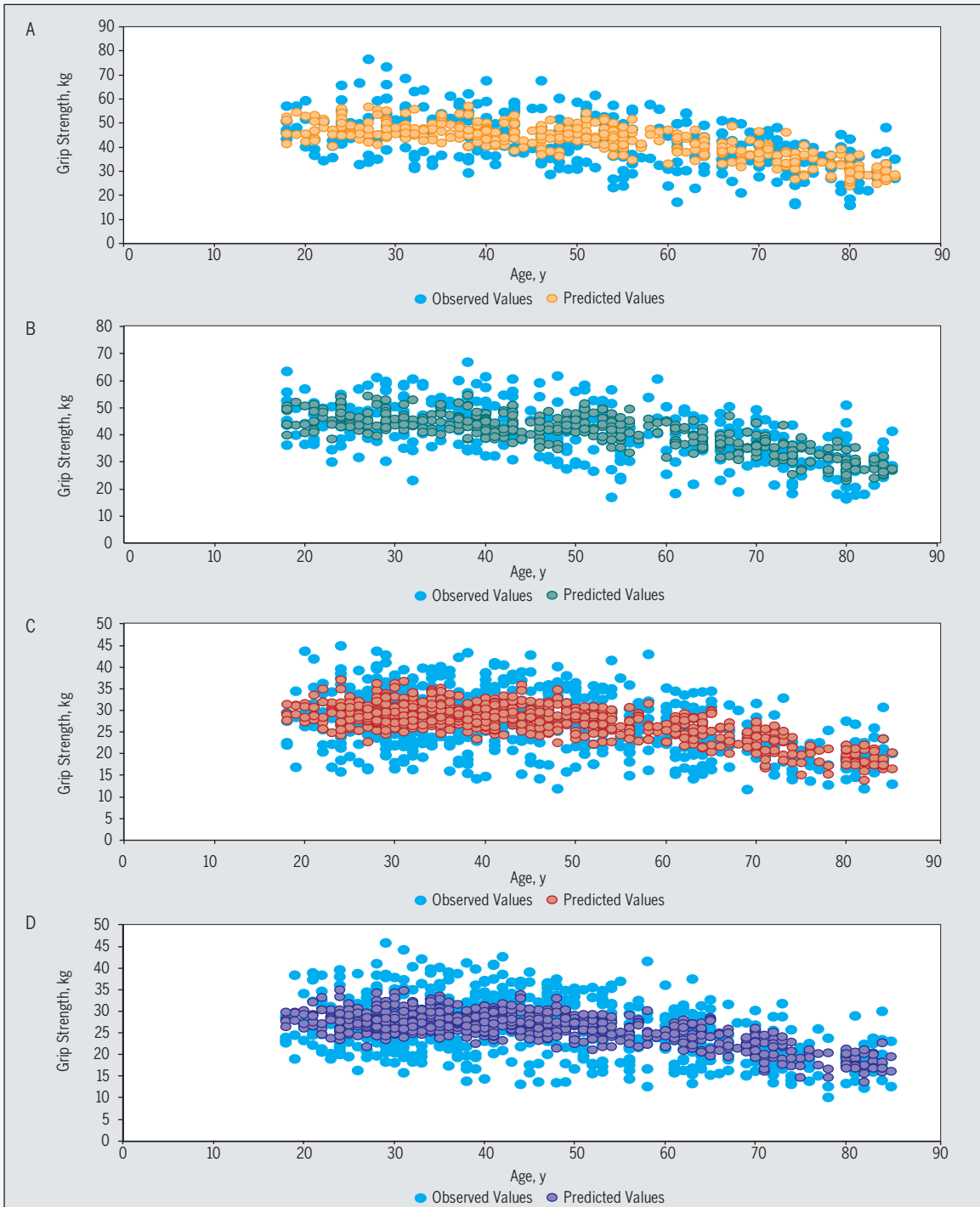
APPENDIX A

Hand/Sex/Age, y	Strength, lb*	Percentile				
		10	25	50	75	90
Female						
18-24 (n = 54)	58.6 ± 14.1	44.1	48.1	54.5	68.3	82.9
25-29 (n = 102)	61.5 ± 14.6	45.0	52.2	60.6	70.1	84.2
30-34 (n = 109)	61.1 ± 13.0	43.2	53.1	60.8	67.9	77.6
35-39 (n = 90)	61.7 ± 13.2	43.4	52.2	60.8	70.5	80.2
40-44 (n = 88)	63.7 ± 14.1	47.8	55.6	64.6	74.1	81.1
45-49 (n = 52)	60.4 ± 15.4	37.7	50.3	59.3	73.6	80.5
50-54 (n = 65)	58.4 ± 14.3	39.0	49.2	58.2	70.3	76.7
55-59 (n = 30)	52.0 ± 14.1	32.2	40.6	51.8	62.2	68.6
60-64 (n = 58)	50.5 ± 13.9	34.8	38.8	49.8	62.2	67.5
65-69 (n = 29)	46.3 ± 14.6	33.1	35.7	47.2	56.9	67.5
70-74 (n = 43)	44.5 ± 12.1	30.2	36.8	46.1	51.8	61.7
75-79 (n = 17)	41.2 ± 12.8	23.6	31.7	41.0	48.5	60.4
80-85 (n = 46)	42.8 ± 8.8	30.6	38.1	42.5	46.3	54.0
*Values are mean ± SD.						

APPENDIX B



APPENDIX C



Scatter plots of the observed and predicted grip-strength values for (A) the dominant hand of men, (B) the nondominant hand of men, (C) the dominant hand of women, and (D) the nondominant hand of women.